

NORMAN WELLS CS, NT, Canada

WMO: 714800

Lat: 65.287N

Lon: 126.753W

Elev: 94

StdP: 100.21

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -42.7	(c) -39.7	(d) -46.7	(e) 0.0	(f) -42.0	(g) -43.7	(h) 0.1	(i) -39.3	(j) 9.5	(k) -15.0	(l) 8.2	(m) -16.1	(n) 0.5	(o) 100	(p) 0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	28.0	16.0	26.0	15.2	24.2	14.5	17.1	25.3	16.3	23.7	15.5	22.3	3.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.2	10.2	18.7	13.4	9.7	18.2	12.4	9.1	17.8	48.4	25.3	46.1	23.7	43.8	22.5	20.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.3	(b) 6.3	(c) 5.5	(e) -44.5	(f) 31.2	(g) 2.4	(h) 1.4	(i) -46.2	(j) 32.3	(k) -47.6	(l) 33.1	(m) -48.9	(n) 33.9	(o) -50.6	(p) 35.0		
(a) 7.3	(b) 6.3	(c) 5.5	(e) -44.2	(f) 18.4	(g) 2.1	(h) 1.0	(i) -45.7	(j) 19.2	(k) -46.9	(l) 19.7	(m) -48.1	(n) 20.3	(o) -49.6	(p) 21.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.8	-24.7	-22.1	-18.2	-5.7	7.3	14.4	16.1	12.9	6.0	-3.6	-18.0	-23.4
	DBStd	16.49	8.47	7.85	8.26	7.14	5.55	4.36	3.44	4.08	4.30	5.04	6.66	7.75
	HDD10.0	5902	1077	898	875	472	120	9	2	17	134	422	841	1036
	HDD18.3	8491	1336	1132	1133	722	344	132	84	173	371	680	1091	1294
	CDD10.0	491	0	0	0	1	35	142	192	108	13	0	0	0
	CDD18.3	36	0	0	0	0	1	14	16	6	0	0	0	0
	CDH23.3	581	0	0	0	0	35	199	241	104	2	0	0	0
	CDH26.7	121	0	0	0	0	5	44	51	21	0	0	0	0
Wind	WSAvg	2.2	1.8	1.9	2.1	2.7	2.7	2.8	2.4	2.4	2.4	2.1	1.8	1.7
Precipitation	PrecAvg	332	20	17	12	15	20	41	56	57	30	30	18	20
	PrecMax	514	62	72	38	63	65	103	117	116	86	87	39	52
	PrecMin	186	0	0	1	0	1	7	12	2	6	6	3	4
	PrecStd	76	13	15	9	12	13	21	26	31	15	18	9	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.1	-1.1	6.5	16.7	27.1	30.5	30.6	29.2	22.2	9.5	-1.4	-1.6
		MCWB	-1.3	-2.8	1.8	7.8	13.5	16.8	17.2	17.4	12.4	4.7	-2.0	-3.0
	2%	DB	-5.9	-5.3	2.6	11.9	23.8	27.8	27.9	26.2	19.2	6.9	-5.3	-7.1
		MCWB	-6.7	-6.6	-0.8	5.4	12.2	15.5	16.5	15.8	11.5	4.0	-6.0	-7.8
	5%	DB	-9.5	-8.5	-1.4	8.1	21.2	25.6	26.0	23.7	16.5	5.0	-6.7	-9.8
		MCWB	-10.0	-9.4	-3.7	3.1	11.3	14.5	15.5	15.1	10.1	3.0	-7.4	-10.4
	10%	DB	-13.5	-11.0	-5.8	5.4	18.1	23.6	24.1	21.3	14.1	3.2	-8.6	-12.7
		MCWB	-13.9	-11.7	-7.4	1.5	9.5	13.8	15.0	14.1	9.2	1.7	-9.2	-13.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.0	-2.6	2.6	7.8	13.9	17.7	18.2	18.1	13.5	6.0	-1.9	-3.2
		MCDB	0.0	-1.2	6.5	16.4	26.0	27.2	26.1	26.7	20.3	8.6	-1.2	-1.7
	2%	WB	-6.7	-6.4	-0.7	5.6	12.5	16.5	17.2	16.8	11.9	4.2	-5.9	-7.7
		MCDB	-6.1	-5.1	2.2	11.5	22.6	25.3	24.9	24.6	17.6	6.1	-5.3	-7.2
	5%	WB	-9.9	-9.3	-3.9	3.3	11.3	15.5	16.6	15.7	10.7	3.0	-7.3	-10.3
		MCDB	-9.4	-8.5	-1.3	7.9	20.0	23.6	23.8	22.0	15.7	5.1	-6.8	-9.8
	10%	WB	-13.9	-11.7	-7.4	1.5	10.0	14.4	15.8	14.6	9.5	1.7	-9.2	-13.1
		MCDB	-13.5	-11.0	-5.9	5.3	17.4	21.7	22.6	19.9	13.7	3.1	-8.6	-12.7
Mean Daily Temperature Range	5% DB	MDBR	8.1	10.1	12.9	12.4	12.9	12.7	12.5	11.4	10.4	7.0	8.0	8.1
		MCDBR	11.7	12.0	15.4	13.9	18.3	15.7	16.2	16.0	15.9	9.4	8.6	10.3
		MCWBR	10.9	11.0	12.0	8.2	9.0	6.7	7.1	7.6	9.2	6.8	8.2	9.8
	5% WB	MCDBR	11.6	12.2	15.0	13.2	17.2	14.1	13.9	13.7	14.3	9.0	8.6	10.2
		MCWBR	10.8	11.2	11.7	7.8	8.6	6.3	6.6	6.9	8.5	6.8	8.2	9.7
Clear-Sky Solar Irradiance	taub		0.156	0.233	0.257	0.317	0.350	0.361	0.384	0.359	0.301	0.255	0.166	0.158
	taud		1.948	2.220	2.253	2.188	2.346	2.393	2.364	2.438	2.548	2.411	1.954	1.951
	Ebn at Noon		392	687	834	846	854	852	815	803	792	662	342	110
	Edh at Noon		20	51	79	110	105	103	103	87	62	42	19	6
All-Sky Solar Radiation	RadAvg		0.13	0.78	2.46	4.59	5.50	5.82	4.96	3.69	2.24	0.85	0.21	0.03
	RadStd		0.01	0.06	0.13	0.17	0.40	0.44	0.31	0.34	0.19	0.07	0.01	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+0.74	+1.50	+1.39	N/A	N/A	N/A	-236	-231	N/A	N/A

Nomenclature: See separate page